

LUZERN, Switzerland

WMO: 066500

Lat: 47.0364N

Lon: 8.3008E

Elev: 1496

StdP: 13.92

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.7	(c) 22.4	(d) 12.8	(e) 11.2	(f) 21.5	(g) 16.3	(h) 13.2	(i) 24.2	(j) 22.7	(k) 48.2	(l) 18.8	(m) 47.9	(n) 4.0	(o) 270	(p) 0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.6	(c) 86.6	(d) 68.8	(e) 83.3	(f) 67.7	(g) 80.0	(h) 66.4	(i) 70.2	(j) 83.1	(k) 68.7	(l) 80.9	(m) 67.3	(n) 78.2	(o) 5.0	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	100.3	75.8	64.4	95.7	74.3	63.2	91.5	72.8	35.1	83.5	33.8	81.2	32.6	78.4	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.0	(b) 12.0	(c) 9.3	(e) 14.9	(f) 91.6	(g) 4.7	(h) 2.2	(i) 11.5	(j) 93.2	(k) 8.8	(l) 94.5	(m) 6.2	(n) 95.8	(o) 2.7	(p) 97.4		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			14.0	73.6	4.7	1.6	10.6	74.7	7.9	75.6	5.2	76.5	1.8	77.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	34.5	35.4	43.1	51.1	57.5	65.0	68.0	66.7	59.5	51.9	41.9	35.5	(6)
(7)		DBStd	13.68	6.85	7.12	7.22	6.81	6.62	6.76	5.71	5.74	5.76	6.35	6.03	6.25	(7)
(8)		HDD50	1961	483	408	230	69	12	1	0	0	3	57	249	449	(8)
(9)		HDD65	5476	947	828	678	419	245	83	35	48	180	407	692	914	(9)
(10)		CDD50	2304	1	0	17	101	245	452	559	518	289	115	7	0	(10)
(11)		CDD65	342	0	0	0	0	13	84	128	101	16	0	0	0	(11)
(12)		CDH74	2609	0	0	0	16	135	698	972	709	79	0	0	0	(12)
(13)		CDH80	706	0	0	0	1	18	199	282	203	3	0	0	0	(13)
(14)	Wind	WSAvg	3.3	3.5	3.5	3.9	3.8	3.7	3.6	3.5	3.1	2.8	2.7	2.8	3.1	(14)
(15)	Precipitation	PrecAvg	59.4	3.2	3.5	3.7	4.6	6.5	6.9	6.9	4.9	4.1	4.2	4.1		(15)
(16)		PrecMax	73.5	7.1	8.5	9.4	9.3	11.3	10.2	13.4	13.6	9.4	7.1	9.8	9.1	(16)
(17)		PrecMin	46.2	0.4	1.4	1.3	0.8	1.5	3.7	2.8	1.7	1.8	0.4	0.0	0.1	(17)
(18)		PrecStd	6.8	1.8	1.8	1.9	2.1	2.6	1.7	2.7	2.7	1.9	1.6	2.1	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.9	57.7	66.5	76.6	82.4	89.7	90.8	90.5	79.9	70.6	61.4	53.5	(19)
(20)			MCWB	49.5	48.1	51.8	57.8	64.2	69.7	70.0	68.6	66.6	60.5	52.9	47.1	(20)
(21)		2%	DB	51.2	52.6	62.6	72.1	77.7	85.4	86.6	85.2	76.4	66.8	56.5	49.2	(21)
(22)			MCWB	45.0	44.4	50.2	56.1	61.9	68.4	68.4	69.3	65.0	58.7	50.6	44.9	(22)
(23)		5%	DB	47.0	48.9	59.1	68.0	73.9	81.8	83.0	81.4	72.5	63.7	52.9	46.4	(23)
(24)			MCWB	42.7	42.5	48.7	54.1	59.9	66.8	67.5	68.1	63.1	57.1	48.9	43.3	(24)
(25)		10%	DB	43.9	45.4	55.0	64.0	70.1	77.8	79.5	77.6	69.1	60.7	50.3	44.1	(25)
(26)			MCWB	40.6	40.6	46.6	52.2	58.1	65.1	66.1	66.4	61.4	56.0	47.0	41.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	48.3	53.3	59.4	66.1	72.3	72.0	71.8	68.2	62.1	53.7	48.2	(27)
(28)			MCDWB	58.0	55.8	63.5	73.6	79.8	85.9	86.9	84.2	77.4	68.6	59.5	51.7	(28)
(29)		2%	WB	45.9	45.5	51.1	56.9	63.7	69.6	69.9	70.0	65.7	59.8	51.5	45.6	(29)
(30)			MCDWB	50.0	51.5	60.4	69.3	75.1	82.5	83.2	82.5	74.4	64.9	55.6	48.4	(30)
(31)		5%	WB	43.1	43.0	49.3	54.9	61.2	67.7	68.4	68.5	63.8	58.0	49.4	43.7	(31)
(32)			MCDWB	46.7	48.3	57.8	66.3	70.9	79.4	80.7	80.0	71.3	62.4	52.4	46.1	(32)
(33)		10%	WB	40.8	40.8	47.3	52.9	59.2	66.0	66.9	66.9	62.0	56.3	47.4	41.7	(33)
(34)			MCDWB	43.5	44.9	54.5	62.4	67.9	76.0	77.7	76.5	68.0	60.1	50.0	44.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.2	11.7	15.9	17.7	16.9	17.3	17.6	16.6	14.7	12.6	10.1	8.5	(35)
(36)			MCDBR	14.9	18.8	23.0	25.3	24.1	24.2	23.6	22.7	19.9	17.5	14.6	12.6	(36)
(37)			MCWBR	10.1	11.8	12.8	12.4	11.7	10.9	10.1	10.1	10.2	10.4	9.4	8.6	(37)
(38)		5% WB	MCDWB	14.4	17.7	21.2	23.5	21.4	22.2	21.6	21.4	18.4	15.8	13.9	11.8	(38)
(39)			MCWBR	9.9	11.5	11.9	11.9	10.9	10.4	9.7	9.9	9.7	9.6	9.3	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.293	0.324	0.372	0.414	0.432	0.437	0.438	0.426	0.395	0.365	0.324	0.290	(40)	
(41)		taud	2.357	2.274	2.212	2.162	2.173	2.215	2.210	2.257	2.322	2.376	2.415	2.380	(41)	
(42)		Ebn at Noon	251	264	267	266	265	264	261	259	257	247	237	238	(42)	
(43)		Edh at Noon	23	31	38	44	45	44	43	40	34	28	22	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	422	703	1085	1391	1560	1817	1778	1548	1183	761	440	348	(44)	
(45)		RadStd	50	92	153	210	154	167	158	148	117	83	59	51	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.61	N/A	N/A	N/A	N/A	N/A
(47) Regional (121 neighbors)		N/A	N/A	N/A	+1.30	+0.72	+0.56	N/A	N/A	+103	+27

Nomenclature: See separate page